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Course: Theory of Computations

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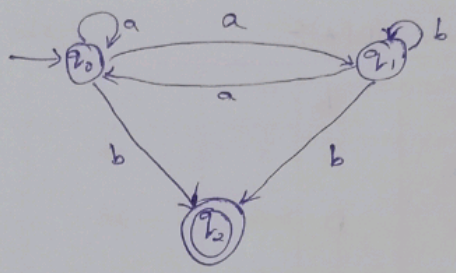
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# CO1 - AT-1

1) Consider the NFA given below which accepts same language L over the alphabet $\Sigma = \{a, b\}$ construct a DFA that represents the same Language L.

*1)




subset construction (DFA):

DFA states:

$A = \{q_0\}$ [start]

$B = \{q_0, q_1\}$

$C = \{q_2\}$ (final)

$D = \{q_1, q_2\}$ final

$\phi = \phi$ trap

Transitions of the given NFA:-

$\delta(q_0, a) = \{q_0, q_1\}$

$\delta(q_0, b) = \{q_2\}$

$\delta(q_1, a) = \{q_0\}$

$\delta(q_1, b) = \{q_1\}$

$\delta(q_0, a) = \phi$

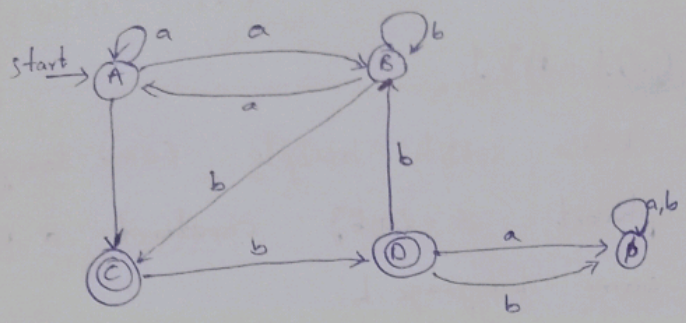
$\delta(q_2, b) = \{q_0, q_1\}$

Start state = $q_0$

final state = $\{q_2\}$

Alphabet = $\{a, b\}$.

State Diagram (DFA):-

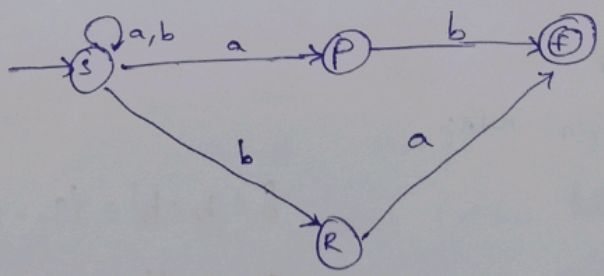



Transition table (DFA):-

| State | a | b |
|---|---|---|
| A | B | C |
| B | B | D |
| C | ∅ | B |
| D | A | B |
| ∅ | ∅ | ∅ |

2) Let L be the language of words of length atleast 2 whose last two letters are different from each other.

A) NFA for L:-




formal definition of NFA

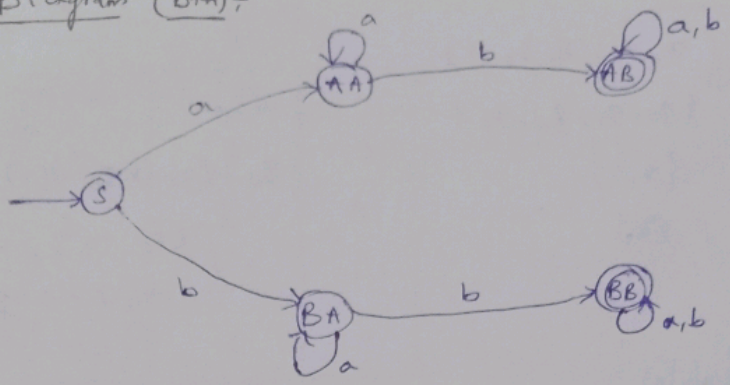
$M = (Q, \Sigma, \delta, S, f)$

$Q = \{S, P, R, f\}$

$\Sigma = \{a, b\}$

$f = \{f\}$

State Diagram (DFA):-

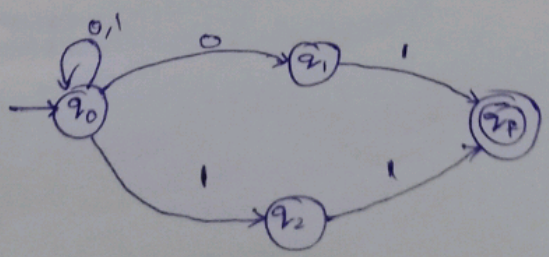



Transition table (DFA):-

| state | a | b |
|---|---|---|
| S | ~~A~~ | B |
| ~~A~~ | ~~AA~~ | ~~AB~~ |
| B | BA | BB |
| ~~AA~~ | ~~AA~~ | ~~AB~~ |
| ~~AB~~ | BA | BB |
| BA | ~~AA~~ | ~~AB~~ |
| BB | BA | BB |

3) Design a ~~NfA~~ that takes a binary string as input and accepts the string only if the last two digits are the same (either 00 or 11).

4) NfA Diagram:-



formal definition:

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{q_0, q_1, q_2, q_f\}$

$\Sigma = \{0, 1\}$

$F = \{q_f\}$

string 01011 is Accepted:

Path:

$q_0 - (0) \rightarrow q_0 - (1) \rightarrow q_0$

$- (0) \rightarrow q_0 - (1) \rightarrow q_2$

$- (1) \rightarrow q_f$

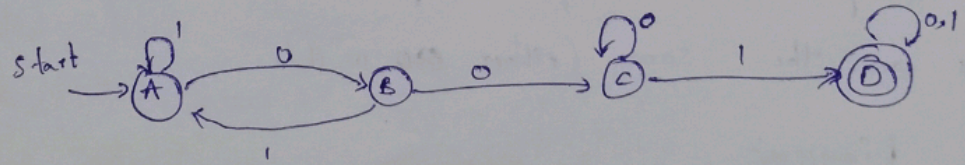
Transition table:

| state | 0 | 1 |
|---|---|---|
| $q_0$ | $\{q_0, q_1\}$ | $\{q_0, q_2\}$ |
| $q_1$ | $\{q_f\}$ | $\phi$ |
| $q_2$ | $\phi$ | $\{q_f\}$ |
| $q_f$ | $\phi$ | $\phi$ |

4) Consider the NFA given below which accepts the language representing the set of strings having 001 as a substring over the alphabet $\Sigma = \{0, 1\}$.

A) NFA to DFA for strings containing substring 001

DFA diagram:

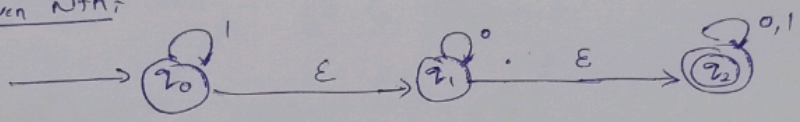


Transition table:-

| state | 0 | 1 |
|---|---|---|
| A | B | A |
| B | C | A |
| C | C | D |
| D | D | D |

5) A NFA with ε-transitions given below find ε-closure for each state and design an equivalent DFA which accepts the same language.

A) Given NFA:-




ε - closures:-

ε - closure $(q_0) = \{q_0, q_1, q_2\}$

ε - closure $(q_1) = \{q_1, q_2\}$

ε - closure $(q_2) = \{q_2\}$

Subset DFA Diagram:-

DFA states (Subsets):-

$A = \{q_0, q_1, q_2\}$ (start, final)

$B = \{q_1, q_2\}$ (final)

$C = \{q_2\}$ (final)

Equivalent DFA Diagram:-




Transition Table:-

| State | 0 | 1 |
|---|---|---|
| A | B | A |
| B | B | C |
| C | C | C |